

Feasibility of pico-hydro and photovoltaic hybrid power systems for remote villages in Cameroon

E.M. Nfah^{a,*}, J.M. Ngundam^b

^a Laboratoire d'Automatique et d'Informatique Appliquée, I.U.T. Fotso Victor, P.O. Box 134, Bandjoun, University of Dschang, Cameroon

^b Automation and Control Laboratory, School of Engineering, P.O. Box 8390, University of Yaounde I, Cameroon

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ABSTRACT

Pico-hydro (pH) and photovoltaic (PV) hybrid systems incorporating a biogas generator have been simulated for remote villages in Cameroon using a load of 73 kWh/day and 8.3 kWp. Renewable energy systems were simulated using HOMER, the load profile of a hostel in Cameroon, the solar insolation of Garoua and the flow of river Mungo. For a 40% increase in the cost of imported power system components, the cost of energy was found to be either 0.352 €/kWh for a 5 kW pico-hydro generator with 72 kWh storage or 0.396 €/kWh for a 3 kWp photovoltaic generator with 36 kWh storage. These energy costs were obtained with a biomass resource cost of 25 €/tonne. The pH and PV hybrid systems both required the parallel operation of a 3.3 kW battery inverter with a 10 kW biogas generator. The pH/biogas/battery systems simulated for villages located in the south of Cameroon with a flow rate of at least 92 l/s produced lower energy costs than PV/biogas/battery systems simulated for villages in the north of Cameroon with an insolation level of at least 5.55 kWh/m²/day. For a single-wire grid extension cost of 5000 €/km, operation and maintenance costs of 125 €/yr/km and a grid power price of 0.1 €/kWh, the breakeven grid extension distances were found to be 12.9 km for pH/biogas/battery systems and 15.2 km for PV/biogas/battery systems respectively. Investments in biogas based renewable energy systems could thus be considered in the National Energy Action Plan of Cameroon for the supply of energy to key sectors involved in poverty alleviation.

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1. Introduction

Cameroon is a Central African country with a surface area of 475,440 km² and a population estimated at 15.5 million inhabitants in 2001. The country possesses the second hydroelectric potential (294 TWh) in Africa after the Democratic Republic of Congo (about 1000 TWh). In 2002, electricity generation from hydro resources accounted for 85% and thermal electricity 15% [1,2]. Existing independent grids in the south, east and north of the country supply electricity mainly to provincial and divisional capitals [1,2], while most sub-divisional and district capitals run on independent mini-grids supported by thermal stations with power capacities in the range of 24 kW–6.4 MW [3].

In spite of the substantial availability of hydroelectric resources in Cameroon, electricity supply remains a serious problem as evidenced by frequent power cuts during the months of January–June since 1999. The output from backup thermal stations during these months of severe drought is insufficient and power rationing no longer guarantees regular operation of industries especially those

connected to low voltage networks. Industries that would like to develop but cannot have access to electricity are just not considered [1,2].

The average rate of electricity access for Cameroonian households in urban and rural areas has been estimated from surveys of household living standards of 1996 and 2001 as 46.1%. This rate comprises 25.9% households connected through utility meters and 20.1% households connected through their neighbours. According to the figures presented by the Rural Electrification Agency at the Libreville conference in 2000, the situation in rural areas is worse. According to those figures, only 1700 out of a total of 9000 councils with populations between 200 and 5000 inhabitants had access to electricity [2]. A recent report in 2002, states that only 40% of city dwellers and 2000 villages out of 30,000 villages in Cameroon have access to electricity. This shows a total access rate for the country of 11% which is quite low in deed [1,2,4,5].

In order to redress the energy crisis specifically in the southern interconnected grid where low water volumes in dams during the dry season affect hydroelectric production, the electricity power corporation and the government have embarked on hydrothermal expansion [1,2,5] which has been proposed in Ref. [6]. Optimal scheduling of large-scale hydrothermal power systems has also

* Corresponding author. Tel.: +237 9688 2160; fax: +237 3301 4601.

E-mail address: em_nfah@yahoo.fr (E.M. Nfah).

been developed in [7] but is yet to be implemented. In spite of the expansion of thermal generation capacities and future development of hydroelectric installations many villages will still remain without electricity for many years to come because of the limited coverage of the nation by existing electric networks and low rate of grid extension [1,2,5].

In 2004, pre-feasibility studies of mini- and micro-hydro sites for rural electrification have identified sites for mini- and micro-hydro installations in the 2–30 MW and 260–1200 kW ranges respectively [2,3]. These power ranges imply that power will be generated at one point and distributed to a number of neighbouring villages since population densities and power demand in most villages are low as evidenced by rural load profiles that have been obtained in some Mekong countries (Thailand, Cambodia and Laos), Uganda and Cameroon [2,8,9]. The development and installation of these micro-hydro power plants will involve much investment and time with the consequence that rural communities will not have access to electricity in future years [2].

In Cameroon, villages at the periphery of sub-urban grid connected areas are often electrified through the extension of medium voltage 30-kV single-wire lines over distances that are often less than 5 km, the installation of autotransformers in the power range of 10–50 kVA and a mini-grid supplying power in a radius in the range of 500–1000 m [1,2]. This practice has produced low electrification rates estimated to be in the range 4–6% for sub-urban and rural areas that harbour 62–67% of the population [1,2,10]. The electrification rate in rural areas is limited by the absence of small-scale decentralised options that could meet the low energy needs of remote villages that often lack income to pay for grid connected service [1,2]. In this regard, the technical and economic feasibility of biomass gasification for power production in Cameroon has been evaluated [10] for decentralised electrification of localities with either wood transformation industries, a population of 3500 inhabitants and a high potential of small industries, or a population of 14,000 inhabitants with high domestic consumption. The energy needs of these localities and the required power generation capacities were estimated at more than 300 MJ/year and 100 kW respectively [10]. However, these decentralised generation options are too large for the over 9000 villages with populations in the range of 200–5000 inhabitants that are not electrified [2,10].

The energy needs of these villages can be met with small-scale off-grid generation options that exploit local renewable energy resources such as pico-hydro, biomass (animal and/or human waste), solar and/or wind energy [1,4,5]. Research efforts for the promotion of renewable energy options that are suitable for village energy needs include the feasibility of wind energy conversion systems for hypothetical villages requiring 70 kWh/day in northern Cameroon [11]. A survey of solar energy applications in the Far North province has also been conducted to establish the contribution of solar energy in the sustainable development of remote villages that are not connected to the northern interconnected grid [12]. The rich solar potential of this region has also been used in modelling of solar/diesel/battery hybrid systems for remote households, clinics and a school [5] as well as in the simulation of off-grid electrification options for remote villages in Cameroon [1].

However, the daily demand of the load profile used in Ref. [1] is too high compared to the daily demand of most villages in Ref. [13]. In Ref. [13], 72 villages out of 83 had daily energy demands less than 60 kWh. The results of the evaluation of the optimal mix of renewable options for these villages showed that one village with the highest biogas availability of 49.5 m³ required four 2 kW biogas generators for its daily energy demand of 44.1 kWh/day [13]. In Ref. [14], a model for the empowerment of rural poor using renewable energy technologies has shown that a 77 m³ biogas digester fed with cow dung from 113 cows could supply biogas for the cooking needs of 61 households estimated at about 45 m³/day.

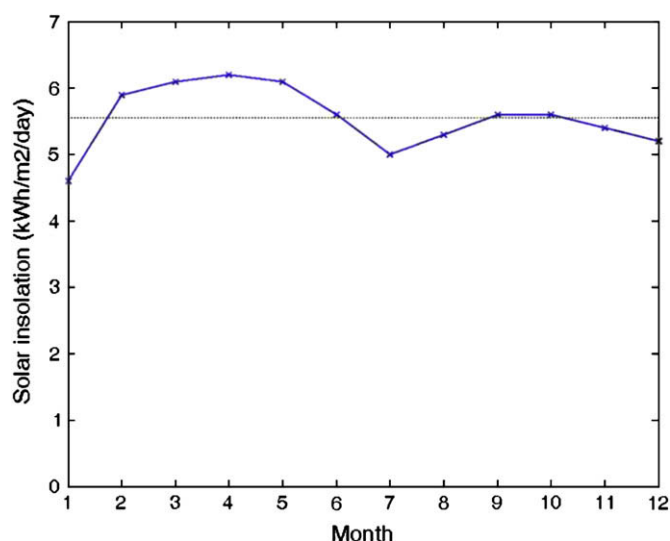


Fig. 1. Monthly solar insolation of Garoua located in North Cameroon [1,2].

The capital, annual operation and maintenance costs of the biogas digester were evaluated in 1988 at US\$ 2817 and US\$ 137 respectively.

In most remote villages' households either use fuel wood, kerosene, automotive batteries or pico-hydro power generation for lighting, listening to the radio and/or watching television [1,8]. Besides household energy needs that can be supplied by solar home systems or automotive batteries that are regularly charged at a battery charging station [15–17] other village needs include quality health care and water pumping. In Cameroon, health centres in remote areas often lack grid electricity, consequently kerosene lamps, candles and/or torches are used during night deliveries. The quality of health care service is limited due to the absence of laboratory equipment and refrigerators for the preservation of vaccines. However, there are a limited number of rural health centres that are equipped with standalone photovoltaic systems or pico-hydro systems. One of these pico-hydro schemes supplies electricity through a standard 2 km 220 V distribution line to a rural health centre located at some 20 km from Bafoussam, the provincial capital of the West Province of Cameroon. The 5 kW pico-hydro scheme in this village is handled by a local management

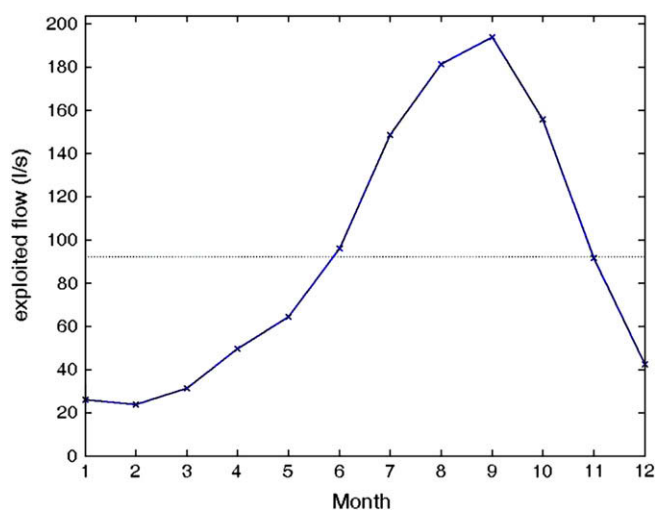


Fig. 2. Exploited flow derived from the flow of River Mungo [1,2].

Table 1

Selected daily load profiles for simulation of renewable energy systems for remote villages in Cameroon [2].

Hour	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	E
Jan	2.19	2.30	2.25	2.81	2.71	5.06	5.52	3.13	2.13	1.88	1.96	1.61	1.73	2.07	2.62	2.48	2.73	4.56	6.37	5.84	5.93	5.11	4.32	2.30	79.7
Feb	2.14	1.98	2.05	1.84	1.91	3.50	6.17	3.98	2.08	2.06	1.91	2.04	2.04	2.25	1.77	2.42	3.96	3.98	6.21	5.64	5.97	4.51	4.51	3.81	78.1
Mar	2.34	2.33	2.18	4.94	2.92	4.76	3.85	2.10	1.55	2.36	1.74	2.12	2.03	2.59	2.32	2.50	2.88	5.47	5.65	5.75	5.41	5.45	5.14	4.43	82.7
Apr	2.40	2.25	2.07	2.06	2.23	2.86	3.56	3.09	1.82	2.36	2.12	2.33	1.64	1.82	2.35	3.26	2.29	3.57	4.40	6.85	5.88	6.36	3.40	2.66	73.6
May	2.62	2.69	2.16	2.32	2.72	3.85	6.87	2.94	2.81	2.44	1.62	2.07	2.28	2.14	2.45	2.50	1.87	2.97	5.33	5.15	5.75	5.27	4.43	3.15	78.7
Jun	2.40	2.25	2.07	2.06	2.23	2.86	3.79	3.59	1.92	1.96	1.96	1.66	1.61	1.73	2.54	2.97	3.98	2.35	4.07	4.49	6.91	5.64	3.82	4.31	72.3
Jul	2.53	2.42	2.02	2.35	2.37	3.38	5.85	3.62	2.38	3.20	3.20	2.44	1.53	1.48	1.84	2.63	3.54	4.26	5.47	5.20	4.95	5.97	3.06	2.62	78.1
Aug	2.54	2.61	1.93	1.67	3.24	4.18	2.30	1.89	1.75	2.33	1.19	1.29	1.61	1.33	1.39	1.25	2.30	5.22	5.33	4.17	4.35	3.88	3.96	2.46	65.2
Sep	2.40	2.12	2.53	3.14	3.08	3.14	3.29	2.71	3.32	1.74	2.12	1.68	1.19	2.60	3.08	3.53	2.64	3.09	3.94	4.17	4.34	5.31	6.77	3.89	75.1
Oct	2.76	2.10	2.10	2.34	2.95	3.02	5.11	4.65	2.34	2.73	2.33	2.55	2.12	2.10	2.25	2.27	2.62	2.67	3.34	4.59	5.18	4.56	4.04	3.46	74.0
Nov	1.75	1.74	1.70	1.70	1.82	2.38	2.64	1.90	1.75	1.74	1.35	1.83	2.20	1.53	1.63	1.70	1.68	3.45	2.70	5.19	4.94	3.32	2.66	2.63	55.5
Dec	2.14	2.02	1.72	1.65	1.97	2.21	2.95	3.19	2.36	2.06	1.86	1.69	2.48	1.82	1.36	1.71	1.74	1.95	3.37	3.46	4.30	3.45	4.01	2.12	57.6
Ave	2.35	2.23	2.07	2.41	2.51	3.43	4.33	3.07	2.18	2.24	1.94	1.94	1.87	1.95	2.13	2.43	2.69	3.63	4.68	5.04	5.32	4.90	4.18	3.15	72.6

committee and was realised at an estimated total project cost of about 10,000€ in 2006. About 10 households have bought electric cables and are effectively connected to the existing low voltage distribution network in spite of the fact that current legislation prohibits this in Cameroon.

The objective of this paper is to simulate pico-hydro and solar options that incorporate a biogas generator for the electrification of remote villages in Cameroon. In this simulation, the scaled annual flow of river Mungo is used for streams in the south of Cameroon, the solar radiation of Garoua is used for villages in the north of Cameroon and the hourly load profile of student's hostel in Cameroon [2] is used as a typical village load profile. Pico-hydro and solar options considered are based on the ac-modular expandable bus configuration and are simulated with HOMER that has been used in Ref. [1].

2. Simulation data

For this simulation, hybrid power systems from the combination of pico-hydro or solar resources with a fuel generator using biogas were considered. Unlike LPG that must be bought in cylinders and transported to most remote villages, biogas can be produced locally using biogas digesters [14,18]. Battery storage is included to reduce the number of start/stop cycles of the backup diesel generator in the hybrid systems simulated as proposed in Ref. [19]. The results obtained in Ref. [13] showed that the energy needs of one village

with the highest biogas potential could be met with only biogas generators consequently, hybrid power options that are simulated include:

- Pico-hydro/biogas generator/battery systems (S1)
- PV/biogas generator/battery systems (S2)

For this simulation, the solar resource of Garoua (9°20' N, 13°23' E) a location in the North Province of Cameroon was selected for remote villages in northern parts of the country. This solar resource was selected as it has been recently used in Ref. [1] for the simulation of off-grid generation options for remote villages in Cameroon. The hydro resource for this simulation was obtained by scaling the annual average flow of River Mungo which has also been used in Ref. [1]. The monthly solar and pico-hydro resource patterns used in this simulation are shown in Figs. 1 and 2. The biomass resource for biogas production was assumed in this simulation to be 350 kg/day from animal waste.

These resource patterns were considered for the simulation of renewable energy options that are suitable for typical village needs for battery charging, refrigeration at rural health centres, water pumping and small commercial activities at a community business centre. To do this, twelve daily profiles with the highest standard deviation in a set of 21 daily profiles [2] were selected for primary load inputs in HOMER from January to December. The single-phase power data selected is shown in Table 1. The energy needs of the

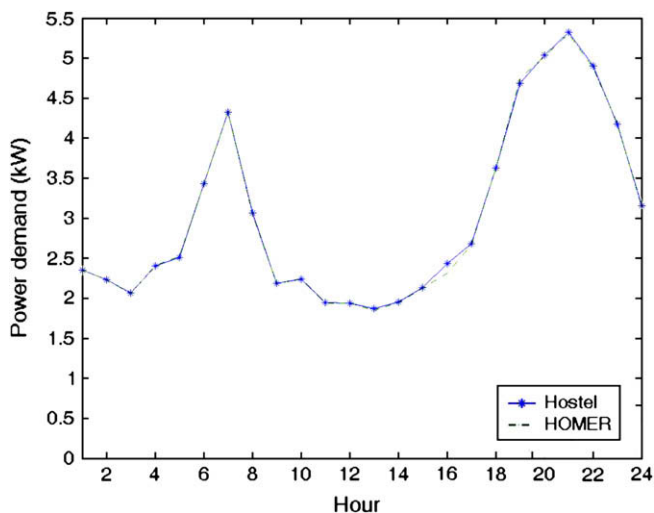


Fig. 3. Hostel average daily load profile and average annual load profile generated with HOMER.

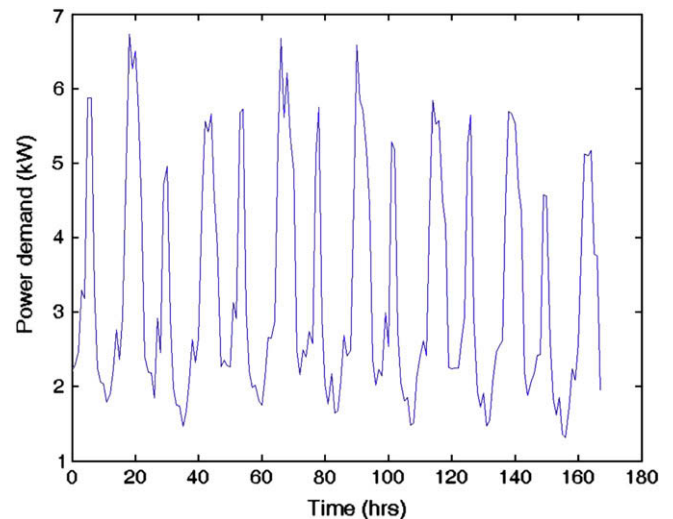


Fig. 4. 168-h load profile of hostel extracted from the annual hourly load data generated with HOMER.

Table 2
Financial data for simulated options.

Description of system parameters	Value
3 kW PV array and string inverter capital and replacement costs (€)	21,000
3.3 kW battery inverter capital and replacement costs (€)	3080
2 V Hoppecke 12 OPzS 1500Ah battery capital and replacement costs (€)	1020
Operation and maintenance cost of 3 kWh battery (€/yr)	50
Operation and maintenance cost of 3 kW PV array (€/yr)	8
10 kW biogas generator capital and replacement costs (€)	6580
Fuel generator operation and maintenance costs (€/h)	0.60
Capital costs of 5 kW pico-hydro generator + electronic circuitry (€)	2500
Replacement cost of 5 kW pico-hydro generator + electronic circuitry (€)	1500
Pico-hydro operation and maintenance costs (€/yr)	1000
Pico-hydro fixed system cost (civil works + 2 km 220 V distribution line)	7500
Pico-hydro system fixed operation and maintenance costs	1000
PV fixed system capital cost (€)	1000
PV system fixed capital operation and maintenance costs (€/yr)	0
77 m ³ biogas digester capital costs (€)	2600
Biogas digester operation and maintenance costs (€/yr)	125
System shortage penalty (€/kWh)	0
Carbon dioxide emissions penalty (€/t)	1.5
Annual real interest rates (%)	6
Project lifetime (yrs)	25
Biogas generator lifetime or operating hours (hrs)	30,000
Cost of grid extension (€/km)	5000
Operation and maintenance cost of grid line (€/yr/km)	125
Grid power price in Cameroon (€/kWh)	0.10

student's hostel is supplied by an auto transformer rated 17.32 kV, 25 kVA supplied by a single wire 30 kV line over a distance of about 2 km.

Electrical energy supplied to the hostel was generally used for operating electric irons, water heaters, electric cookers in addition to lights and radio cassettes but rarely television. The daily energy needs of the hostel over the three weeks measurement period was found to be in the range 44.2–84.6 kWh and the peak daily power demands were 3.2–6.9 kW [2]. This energy demand is comparable to that of two Senegalese villages each of 2000 inhabitants electrified with a single-phase mini-grid (220 V, 50 Hz) [1,20]. A careful investigation of the energy needs of the two villages each located at a little more than 15 km from the national grid had resulted in the design of two standalone photovoltaic systems for an estimated village daily energy consumption of 70.7 kWh and 60 kWh with an average of 0.37 kWh/day per household. These systems were installed in 1989/1990 and upgraded to 26.64 kWp and 20 kWp in 1995/1996 to solve problems of overloading [1,20]. The daily energy demand of the hostel is also comparable to the 70 kWh demand that has been considered in Ref. [11].

The average daily power and energy demand derived from the 12 selected daily load profiles of the hostel are included in Table 1. The average daily load profile obtained is shown in Fig. 3 as well as the annual hourly average derived from the annual hourly load data generated with HOMER for hourly and daily noise inputs set to 5%. Fig. 3 shows that there are morning and evening peaks at 7 a.m. and 9 p.m. respectively. Morning and evening peaks have also been observed at 7 a.m. and 8 p.m. in the daily load profile of a village in Laos with a daily energy demand of 33.4 kWh [8]. Similarly morning and evening peaks have been encountered in the load profile of a village in Uganda and in the load profile that has been constructed for urban households in Cameroon [9,21]. Consequently, the selected load profile of the hostel is suitable for the simulation of off-grid electricity generation options for remote villages.

The load profile generated with HOMER has a daily energy demand of 73 kWh/day and 8.3 kWp. Thus a 10 kW biogas generator was selected for meeting the peak load and possible growth in energy demand. In the absence of a fuel generator, the battery inverters must be of sufficient capacity to meet the peak power demand. The one-week load profile generated with HOMER is shown in Fig. 4.

In order that HOMER should compute the levelized life cycle costs of electricity for the different renewable energy options under consideration, current financial information for power components in the different options must be specified. In this simulation the capital costs of the pico-hydro system (water turbine + generator + electronic circuitry) and the fixed capital costs (project design + civil works + 2 km low voltage distribution line + project delivery) were estimated from project cost distributions of two pico-hydro schemes in Kenya [22] as 25% and 75% respectively. Using the estimated project cost of 10,000€, the capital, replacement, operation and maintenance costs of the pico-hydro system were estimated at 2500 €, 2000 € and 1000 €/year respectively. Using the generator sizes and prices stated in Ref. [1,8], the cost of a 10 kW diesel generator on European markets was estimated using linear interpolation at 470 €/kW. Applying a 40% price increase in local markets, the cost of a 10 kW biogas powered generator was evaluated at 6580€. Single-wire grid extension costs, annual grid operation, maintenance costs and the grid power price used in Ref. [1] are also used in the simulation. Using the costs of the 77 m³ biogas digester realised in Ref. [14] with a 200% increase, the capital, operation and maintenance costs of the biogas plant were estimated at 2600€ and 125€ respectively.

Using prices from retailers for European markets, the capital and replacement cost of a 3 kW photovoltaic generator (3 kW photovoltaic array plus 3 kW string inverter) were estimated with a 40% increase at 21,000€ while that of a 3.3 kW bidirectional inverter was evaluated similarly as 3080€. The same capital and replacement costs of 2 V Hoppecke 12 OPzS 1500 vented lead acid tubular plate batteries rated 3 kWh used in Ref. [1] are maintained as well as the same operation and maintenance cost of the 3 kW photovoltaic array and 3 kW battery storage. The round trip efficiency of the selected battery is 82% and its minimum state of charge is 30%. The efficiency of the bidirectional inverter is 90% while that of the PV-string inverter is assumed to be 90% as in Ref. [8]. The financial data used in this simulation are shown in Table 2.

3. Hybrid system optimization procedure

As a consequence of the 24 V dc-input voltage of the selected bidirectional inverter, batteries that were considered in the battery module were 12, 24 and 36. In the converter module, total power converter capacity ratings of 3.3, 6.6 and 9.9 kW were selected. In the PV generator module five values for PV capacity 3, 6, 9, 12, 15 kW were considered. In the hydro module, the minimum and maximum exploitable flows for the micro-hydro scheme were set to 50% and 100% of the nominal flow respectively. Using data of the selected water turbine for an electrical output of 5 kW, the available head was set to 10 m, the design flow to 92.6 l/s and the turbine-generator efficiency was set to 55%. In the head loss module, the steel pipe diameter was set to 250 mm and the length of the pipe to 15 m and the head loss was evaluated as 1.72%. Since the rating of the biogas generator is less than 12 kW, the minimum load ratio

Table 3
AC-bus configurations, energy costs, breakeven grid distances with a biomass resource cost of 25 €/tonne.

Option	PV [kW]	pH [kW]	G [kW]	B [kWh]	BC [kW]	C [€]	NPC [€]	COE [€/kWh]	RF [%]	m [tonne]	h [hrs]	E [%]	d [km]
S1	0	5	10	72	3.3	46740	119180	0.352	100	43	1423	25	12.9
S2	3	0	10	36	3.3	46500	133765	0.395	100	94	3239	5	15.2

Table 4

Cost of energy and breakeven grid distances for hybrid options with biogas generator.

Biomass cost (€/t)		0	5	10	15	20	25
COE (€/kWh)	S1	0.260	0.268	0.277	0.285	0.344	0.352
	S2	0.292	0.315	0.337	0.360	0.378	0.396
<i>d</i> (km)	S1	8.2	8.6	9.1	9.5	12.5	12.9
	S2	9.9	11.0	12.2	13.3	14.2	15.2

was set to 70% in the generator modules as adopted in Refs. [1,2,4,5].

In the generator control module, cycle charging and a state set point of charge of 50% were selected. In the constraints module the maximum annual capacity shortage was set to 0% while the minimum renewable fraction was set to 60%. This renewable energy fraction was selected since renewable energy fractions greater than 60% have been obtained in PV hybrid systems that have been simulated for villages in Mekong countries [8]. An hourly operating reserve of 10% was specified to ensure that the system can cope with a 10% increase in the hourly primary load. Other relevant parameters in the economics, generator control, carbon emissions, and constraints modules were also specified.

The effect of biomass (cow dung, piggery and poultry waste) collection and transportation cost is investigated with: 0 €/tonne, 5 €/tonne, 10 €/tonne, 15 €/tonne, 20 €/tonne and 25 €/tonne.

4. Results

The results obtained with a biomass resource cost of 25 €/tonne are shown in Table 3. In Table 3, PV represents PV array power, pH pico-hydro generator power, B battery storage, G biogas generator power, BC bidirectional converter power, C capital costs of power system, NPC net production costs, COE cost of energy, RF renewable energy fraction, *m* mass of biomass, *h* biogas generator hours, *E* percentage of excess energy and *d* breakeven grid distance.

It is observed in Table 3, that the size of the hydro and photovoltaic generator required in the hybrid system is quite small in areas with high biogas availability. The variations of the cost of energy and breakeven grid distances with the changes in the cost of biomass resource are presented in Table 4 and illustrated in Figs. 5 and 6. It is noted from these graphical illustrations that the cost of

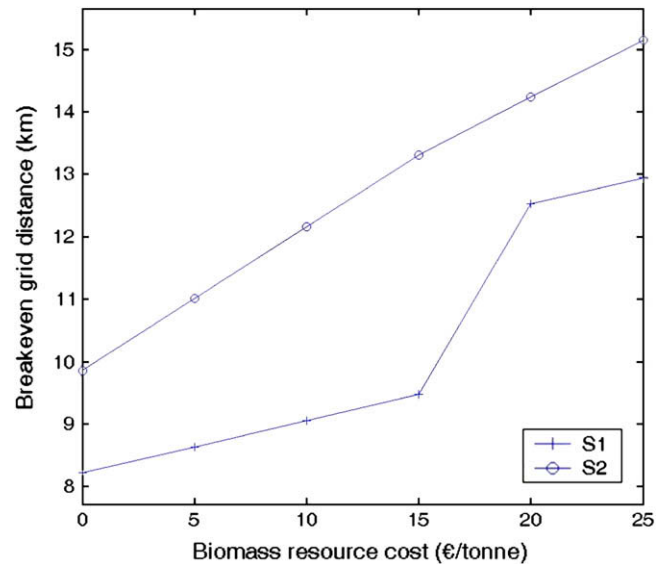


Fig. 6. Breakeven grid distance for hybrid options with biogas generator.

energy and breakeven distances reduce with a reduction in the cost of biomass resource.

For a biomass resource cost of 25 €/tonne, the energy costs and the breakeven grid distances for simulated options were found to be:

- 0.352 €/kWh and 12.9 km respectively for option S1;
- 0.396 €/kWh and 15.2 km respectively for option S2.

These off-grid options compete favourably with grid extension and so could be used in the in the current National Energy Action Plan for the provision of energy services to remote villages, schools, rural health centres and rural businesses that are often located at distances outside the breakeven distances that have been computed. These off-grid options can be used to reduce the number of villages that currently lack grid power supplies and also the national poverty level of 40.2 [1,2].

5. Conclusion

Pico-hydro and photovoltaic hybrid systems incorporating a biogas generator for the electrification of remote villages in Cameroon have been simulated with a hostel load profile with an energy demand of 73 kWh/day and 8.3 kWp. The results of this simulation showed that for a biomass resource cost of 25 €/tonne, a 5 kW pico-hydro plant with a 10 kW biogas generator and 72 kWh battery storage produced electricity at a cost of 0.352 €/kWh for remote villages with an annual pico-hydro flow of 92 l/s, a head of 10 m, a penstock of length of 15 m and diameter of 250 mm. It was also found that a 3 kWp photovoltaic generator with a 10 kW biogas generator and 36 kWh battery storage produced electricity at a cost of 0.395 €/kWh for remote areas with an average annual insolation of at least 5.55 kWh/m²/day. Furthermore, a reduction in the cost of biomass resource resulted in reduced energy costs and breakeven grid distances.

These results show that off-grid options based on renewable energy resources could be a suitable alternative for rural electrification in the low power range (10–50 kW) in Cameroon. Thus electricity needs of most remote villages in Cameroon can be met with renewable energy options that include biogas generators with local production of biogas. The use of biogas results in lower electricity generation costs and excess biogas can be used for domestic thermal needs as has been done elsewhere [14,18].

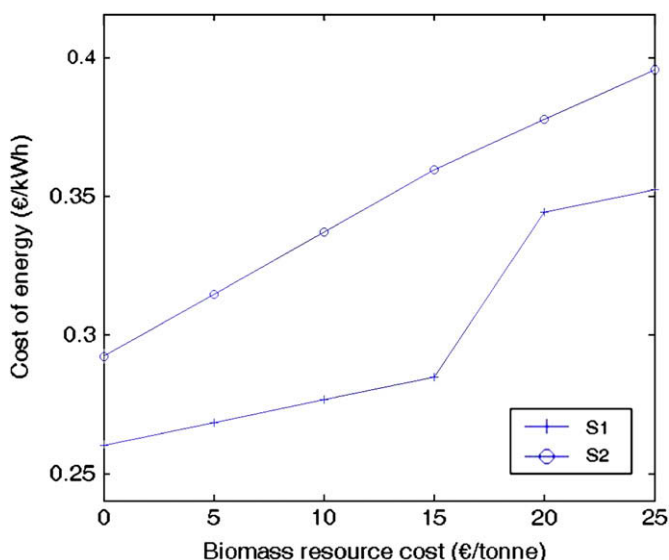


Fig. 5. Cost of energy for hybrid options with biogas generator.

In order to deploy biogas-based renewable energy options in Cameroon, the availability of biogas in remote villages needs to be evaluated as in Ref. [13]. There is also the need to create databases for the solar and pico-hydro potential of remote villages that would be used in the simulation of biogas-based renewable energy options. A study of the energy needs of remote villages in relation with the socio-economic status of rural households is also required. This site-specific information is important for the survival of systems that would be sized, simulated and installed in remote communities in Cameroon [1].

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